SEATRACE FRONTEND: TECHNOLOGY USAGE REPORT

Verified inventory of every library, browser API and build tool used in marine-ai/frontend

Smart India Hackathon (SIH 2026) · Ministry of Earth Sciences (MoES)

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Date: September 7, 2026	**Basis:** package-lock.json plus static analysis of src/ and dist/

1. Summary

This report documents every technology actually present in the SeaTrace frontend (marine-ai/frontend). It is an inventory of what is installed and imported in source, not a wish-list: each dependency below was resolved from package-lock.json and each usage claim was verified by searching src/. The application is a single-page React 19 client that renders the public 'GhostNet' editorial experience and a tactical sonar console from one bundle, and it runs fully offline against a client-side simulation engine when the Spring Boot backend is unreachable.

2. Runtime Dependencies (shipped to the browser)

Package	Version	Role	How it is actually used
react	19.2.8	runtime	UI runtime. Imported in 32 of 41 source files. Function components only - no class components anywhere in the tree.
react-dom	19.2.8	runtime	DOM renderer. createRoot in src/main.tsx, mounted inside <StrictMode>.
three	0.185.1	runtime	WebGL engine. Used in exactly 2 files - Globe3D.tsx and RealisticOceanGlobe3D.tsx (937 LOC) - for the rotating textured ocean globe (THREE.WebGLRenderer).
@types/three	0.185.4	runtime	Type definitions for three. Declared as a runtime dependency rather than a devDependency.
lucide-react	1.41.0	runtime	Icon system. The most widely imported third-party package: 29 of 41 files pull icons from it.
framer-motion	13.2.0	runtime	Animation library, used in ONE file: components/ui/number-ticker.tsx (useSpring, useMotionValue, useInView, useReducedMotion). All other motion in the app is hand-written CSS and requestAnimationFrame.

3. Build, Styling and Quality Toolchain (dev only)

Package	Version	Role	How it is actually used
vite	8.2.2	build	Dev server (port 5173) and production bundler. Configured with an /api proxy to http://localhost:8080 that returns a synthetic 503 JSON body when the backend is down.
@vitejs/plugin-react	6.1.1	build	React Fast Refresh and JSX transform for Vite. Uses the Oxc-based pipeline.
typescript	6.0.3	build	Type checking (tsc -b) ahead of every production build. target es2023, module esnext, moduleResolution bundler, jsx react-jsx, noEmit.
tailwindcss	4.3.3	styling	Utility CSS engine, v4 CSS-first config. No tailwind.config.js exists - the theme lives in an @theme block inside src/index.css.
@tailwindcss/vite	4.3.3	styling	First-party Vite plugin for Tailwind v4; replaces the legacy PostCSS-plugin integration.
postcss	8.5.28	styling	CSS transform pipeline underneath Tailwind.
autoprefixer	10.5.5	styling	Vendor-prefixing of generated CSS.
oxlint	1.81.0	quality	Linter (Rust-based, replaces ESLint). Config .oxlintrc.json enables the react, typescript and oxc plugins; rules-of-hooks is set to error.
playwright	1.63.0	testing	Headless browser driver used by three scripts in frontend/scratch/ that launch MS Edge against the dev server and screenshot animations (hero typewriter, entrance dive, ticker).
@types/node	24.13.3	build	Node type definitions for vite.config.ts.
@types/react	19.2.18	build	React type definitions.
@types/react-dom	19.2.7	build	React DOM type definitions.

4. What Is Deliberately Not Used

Deliberately absent: there is no router (react-router), no state-management library (Redux, Zustand, Jotai), no data-fetching library (React Query, SWR, axios), no component library (MUI, shadcn/ui, Radix), no charting library (Recharts, Chart.js, D3) and no test runner (Vitest, Jest). Routing is a useState view switch seeded from URLSearchParams ('?view=console', '?theme=light', '?assistant=open'); server state is plain fetch() in services/api.ts; every chart in the Explainable AI panel is hand-computed SVG. The full dependency tree is 133 installed packages, which is unusually small for an application of this surface area.

5. Native Browser Platform APIs

Platform API	Where	Purpose
HTML5 Canvas 2D	SonarCanvas.tsx (653 LOC), Globe3D.tsx	Sonar swath rasterisation, pan/zoom, overlay drawing and the 2D fallback globe.
WebGL via three.js	Globe3D.tsx, RealisticOceanGlobe3D.tsx	Two THREE.WebGLRenderer instances driving the textured Earth (7 JPG/PNG textures in public/textures).
Web Audio API	AppEntranceDive.tsx, HeroSection.tsx, RealisticOceanGlobe3D.tsx	18 AudioContext call sites synthesising sonar pings and the ambient dive soundscape - no audio files are shipped.
Web Speech API	AIAssistantModal.tsx (866 LOC)	5 speechSynthesis call sites give the in-app assistant a spoken voice.
requestAnimationFrame	4 call sites across canvas and globe components	Frame loop for the sonar sweep and globe rotation, in place of an animation library.
ResizeObserver	2 call sites	Keeps canvas and WebGL viewports in sync with their containers.
localStorage	App.tsx	Persists the chosen theme under the key 'ghostnet_theme'.
AbortController	services/api.ts	Per-endpoint request timeouts: 1200 ms health, 1500 ms statistics and detections, 3500 ms chat, 8000 ms image analysis.
FormData and fetch	services/api.ts, App.tsx	Multipart upload of a sonar image to POST /api/analyze; 6 fetch call sites in total.
URL.createObjectURL	UploadModal.tsx	Local preview of the uploaded sonar image before analysis.
Dynamic import()	services/api.ts line 107	The 438-LOC conversational engine is code-split and only loaded when the backend chat endpoint fails.

6. Application Architecture

• **Composition:** 41 source files, 11,197 lines of TS/TSX/CSS. 36 components split across two suites: components/ghostnet/ (18 files, the public editorial site) and components/ (17 files, the tactical console), plus components/ui/.
• **State model:** Local React state only - 104 useState, 34 useEffect, 26 useRef, 7 useMemo, 7 useCallback. App.tsx (525 LOC) is the single owner of shared state and passes it down as props. useImperativeHandle exposes SonarCanvasHandle so the parent can drive the canvas.
• **Type system:** src/types/sonar.ts (153 LOC) is the domain contract: RiskLevel, BBox, StageTiming and five signature-detail interfaces (Periodicity, Linearity, Circularity, Angularity, Irregularity) mirroring the backend classification engine.
• **Services layer:** Three files. api.ts (110 LOC) is the network boundary; mockEngine.ts (524 LOC) is a full client-side forward model producing detections, statistics and sample history; chatEngine.ts (438 LOC) is the offline assistant.
• **Offline-first:** Every endpoint follows the same pattern: try the live backend behind an AbortController timeout, fall back silently to the simulation engine, and mark the result isLive:false so the UI can badge it.
• **Styling:** Tailwind v4 utilities plus 528 lines of hand-written CSS (index.css 344, App.css 184). Theming is CSS custom properties (--accent-neon #00ff88, --bg-main, --card-bg and others) swapped by a body.dark / body.light class, with @custom-variant declarations so Tailwind can target them.
• **Typography:** Four Google Fonts loaded from index.html with preconnect: Space Grotesk (brand), Plus Jakarta Sans (display and body), Inter (fallback body), JetBrains Mono (telemetry readouts).

7. Largest Components by Line Count

Component	LOC	Responsibility
ghostnet/RealisticOceanGlobe3D.tsx	937	three.js textured Earth with acoustic resonance audio
ghostnet/AIAssistantModal.tsx	866	In-app assistant UI with speech synthesis
SonarCanvas.tsx	653	Canvas 2D sonar viewer: pan, zoom, overlays, ruler
ghostnet/UploadModal.tsx	545	Sonar image upload, preview and dispatch
App.tsx	525	Root state owner and view switch
services/mockEngine.ts	524	Client-side detection simulation and sample datasets
ghostnet/BeginnerEducationSection.tsx	479	Explanatory ghost-net content for non-experts
services/chatEngine.ts	438	Offline conversational engine (code-split)
ghostnet/SonarAnalysisViewer.tsx	425	Public-facing analysis walkthrough
ghostnet/GlobalOceanMap.tsx	374	Global detection map

8. Build Output and Weight

Artefact	Size	Notes
index-*.js	1,074,471 B (1.02 MB)	Main bundle. three.js and framer-motion dominate this figure.
index-*.css	141,184 B (137.9 KB)	Tailwind output plus the hand-written ocean and sonar CSS.

Artefact	Size	Notes
chatEngine-*.js	14,172 B (13.8 KB)	The one code-split chunk, loaded on demand.
public/ assets	29 MB	7 Earth textures, 11 sample sonar PNGs with paired JSON ground truth, 4 downloadable reports, favicon and icon sprite.
dist/ total	30 MB	Static output - deployable to any CDN or static host; no Node server required.

Build command: tsc -b && vite build. Dev: vite on port 5173 with the /api proxy pointed at the Spring Boot service on port 8080. Lint: oxlint. The proxy has a custom error handler that answers with { error: 'Backend offline', fallback: true } and HTTP 503 instead of a connection reset, so the frontend fallback path is exercised the same way in development as in production.

9. Observations and Recommended Next Steps

• **framer-motion earns 1 file out of 41.** The whole package ships in the main bundle to animate a number ticker. Either replace it with a requestAnimationFrame tween (the codebase already does this everywhere else) or code-split number-ticker.tsx.
• **The bundle is a single 1.02 MB chunk.** three.js is imported statically in two components even though the globe is below the fold. A React.lazy boundary around Globe3D and RealisticOceanGlobe3D would move roughly 600 KB off the critical path.
• **@types/three sits in dependencies.** Type-only packages belong in devDependencies; it does not affect the bundle but it does affect install size.
• **There is no automated test suite.** Playwright is installed, but the three scripts in scratch/ are screenshot probes with hard-coded absolute paths, not assertions. No test runner is configured.
• **public/ is 29 MB of unoptimised assets.** The Earth textures are JPG and PNG; converting them to WebP or AVIF would cut first-load weight substantially for the globe.

All version numbers are the resolved versions from package-lock.json, which differ in places from the semver ranges in package.json (oxlint resolves to 1.81.0, typescript to 6.0.3, @vitejs/plugin-react to 6.1.1, @types/react-dom to 19.2.7).